

Review of Critical Components of Data-Driven Nudge Theory on Enhancing Conversion Rates in Digital Marketing Campaigns

Idha Kristiana

*Computer Science Department
Doctor of Computer Science
Bina Nusantara University
Jakarta, Indonesia 11480
*Corresponding author :
Idha.kristiana@binus.ac.id*

Harjanto Prabowo

*Computer Science Department
Doctor of Computer Science
Bina Nusantara University
Jakarta, Indonesia 11480
harprabowo@binus.ac.id*

Ford Lumban Gaol

*Computer Science Department
Doctor of Computer Science
Bina Nusantara University
Jakarta, Indonesia 11480
fgaol@binus.edu*

Nunung Nurul Qomariyah

*Computer Science Department
School of Computing and Creative Arts
Bina Nusantara University
Jakarta, Indonesia 11480
Bandung, Indonesia
nunung.qomariyah@binus.edu*

Abstract—This study explores the application of nudge theory to enhance conversion rates in digital campaigns, highlighting the gap between its proven effectiveness and suboptimal implementation. Despite its potential, many digital campaigns fail to fully leverage nudge strategies due to a lack of comprehensive frameworks. This systematic literature review (SLR) addresses two key research questions: What are the critical components for effectively integrating data-driven nudges? What approaches and techniques are efficient in utilizing data-driven nudge strategies to improve conversion rates? The review identifies essential components such as data quality, audience understanding, appropriate AI and ML model development, and segmentation and personalization. It also highlights efficient approaches and techniques, including behavioral insights, context-aware nudges, and multi-channel integration. The novelty of this study lies in proposing an integrated framework that systematically addresses these components and techniques, providing a structured strategy for optimizing data-driven nudges. This framework aims to bridge the existing implementation gaps and maximize the effectiveness of nudges in digital campaigns, ultimately enhancing conversion rates.

Keywords—Data Driven Nudges, Digital Marketing Campaigns, Consumer Behavior, Artificial Intelligence

I. INTRODUCTION

A. Data-Driven Nudge

Data-driven nudges have become a potent instrument in numerous fields, including digital marketing campaigns, for influencing user behavior and decision-making. By utilizing data analytics and behavioral insights, these interventions are able to subtly affect user behavior without requiring explicit incentives or constraints [1]. Through a comprehensive comprehension of the fundamental psychological, social, and contextual elements that impact user behavior, data-driven nudge can proficiently direct users towards intended results

while upholding their independence and liberty of selection[2].

Data-driven nudges have garnered considerable interest in numerous domains, including finance, marketing, and public health, in recent times, on account of their capacity to enhance user experiences and results [3]. These factors hold significant relevance within the realm of digital marketing, as organizations consistently strive to enhance their campaigns and bolster conversion rates [4]. Through the implementation of data-driven nudge strategies, organizations can enhance their comprehension of the target demographic and develop marketing interventions that effectively connect with their users, culminating in augmented conversion rates and enhanced customer contentment [5].

B. Digital Marketing Campaigns

Digital marketing campaigns have emerged as a crucial component of contemporary marketing strategies, facilitating the expansion of business reach to a worldwide audience via diverse digital platforms, including but not limited to social media, email, and search engines [6]. The objective of these marketing initiatives is to actively involve users, increase recognition of the brand, and ultimately stimulate desired actions, such as a purchase or newsletter subscription. Nevertheless, the efficacy of these campaigns is profoundly tied to their capacity to transform prospective customers into tangible purchasers. For digital marketing campaigns, conversion rates, which represent the proportion of users who carry out the intended action, are crucial performance indicators [7].

Elevated conversion rates are indicative of a successful campaign that is generating conversions from its intended audience. Conversely, diminished conversion rates indicate that the campaign might require modifications to more effectively connect with its target demographic[8]. Businesses are under constant pressure to optimize their

campaigns and increase their conversion rates to remain competitive in the current digital marketing environment. In order for enterprises to optimize their digital marketing endeavors, it is crucial that they comprehend the variables that impact conversion rates and employ tactics to enhance them[9].

This systematic literature review aims to investigate the essential elements of data-driven nudge frameworks and their potential contribution to the improvement of conversion rates in digital marketing initiatives.

II. THEORITICAL FOUNDATION

A. Nudge Theory

The Nudge Theory, which was initially introduced by Thaler and Sunstein [10], places significant emphasis on the notion of exerting influence over the decisions and behaviors of individuals by means of nuanced modifications to the choice architecture, all while preserving their autonomy. According to this theory, individuals can be influenced to make choices by manipulating the presentation of options or the structuring of information [11]. Nudge Theory is predicated on the utilization of psychological and behavioral economics principles to comprehend the decision-making process of individuals to develop interventions that steer them toward more favorable options without requiring them to perform actions [12]. Nudge Theory is an indispensable framework within the realm of digital marketing for influencing consumer behavior and maximizing conversion rates. By implementing digital marketing strategies in accordance with the tenets of Nudge Theory, organizations could craft user experiences that imperceptibly impact consumer choices, thereby resulting in heightened levels of engagement and conversions. The effective implementation of nudges within digital marketing frameworks has the potential to greatly influence consumer behavior and facilitate the achievement of desired results [13].

B. Behavioral Economics

Behavioral economics (BE) is crucial for comprehending the essential elements of data-driven nudge theory, particularly in the context of improving conversion rates in digital marketing efforts [14]. This discipline, known as behavioral economics, combines principles from psychology and economics to elucidate the reasons behind customers' frequent illogical behavior, which is often driven by cognitive biases and heuristics [15]. Behavioral economics provides valuable insights for marketers to create nudges that are in line with innate human inclinations, such as loss aversion, the endowment effect, and social proof. Marketers can achieve effective influence on consumer behavior and achieve improved conversion rates by comprehending these biases and creating tailored nudges [16].

Behavioral insights are essential in digital marketing initiatives for creating customized and timely prompts that deeply connect with consumers [17]. Utilizing the scarcity bias by offering limited-time deals can induce a sense of urgency, leading to faster purchasing choices. Personalization leverages the familiarity bias, in which customized recommendations create a sense of being understood and appreciated by consumers, thereby enhancing the probability of conversions. By using data-driven nudges based on behavioral economics concepts, marketing tactics

can greatly improve their effectiveness. These nudges are designed to fit with consumers' natural tendencies and prejudices, resulting in increased campaign success [18].

III. METHODOLOGY

A. Objective

The objective of this systematic review is to investigate and consolidate the current body of literature concerning the significance of critical component data-driven nudge in the contemporary marketing environment. This chapter provides an overview of the systematic literature review methodology that was implemented in this study, in accordance with the Kitchenham Guidelines [19].

B. Research Questions

Our primary research questions are:

No	Research Question (RQ)
RQ1	What are the critical components for effectively integrating data-driven nudges?
RQ2	What approaches and techniques are efficient in utilizing data-driven nudge strategies to improve conversion rates?

C. Review Protocol

The process begins by defining Inclusion and Exclusion Criteria to guide the selection of relevant studies. Inclusion criteria specify studies published since 2019, written in English, and focused on nudge theory in digital marketing. Exclusion criteria eliminate non-academic materials and those not directly related to the field. Keywords utilized for the search include "Data Driven Nudges", "Digital Marketing Campaigns", "Nudge Theory" and other related terms. The databases accessed for this research comprised Science Direct, Scopus, AC Digital Library, Springer Link, and IEEE Xplore. Initially, 592 number of articles were identified. After removing duplicates, titles, and abstracts of 81 number articles were screened, leading to a more detailed examination of 50 number articles' full texts. Post this rigorous screening, 50 number articles were finalized for the review.

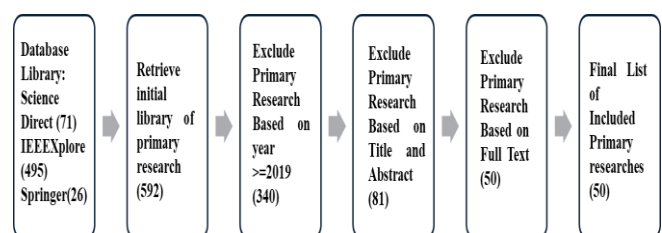


Fig. 1. The SLR Process.

Each chosen article was judged on its quality using set standards, including how clear the research goals were, how strong the technique was, how carefully the data was collected, and how relevant the article was overall. Articles that didn't meet these standards were thrown out. The finished articles' data were carefully taken out and put into an extraction table. The information included the study's goals, the methods it used, its main results, and its most important implications. The gathered data was subjected to thematic analysis, which put the results into groups based on clear themes. This made it easier to see patterns, which led to a more organized review of the literature. The systematic

review results are given in Chapter 4, which also talks about how the literature was gathered and how the research questions were answered. Literature's most important ideas, patterns, and differences are brought to light.

D. Data Extraction and Synthesis

The review started with a screening process that looked at the titles and abstracts of the studies and threw out any that didn't meet the standards. Only those that did were then given a full-text review. A standard way of gathering data was used, with a form being used to get important details like authors, release year, goals, methods, results, and conclusions. A narrative synthesis was used to look at the extracted data and find patterns, differences, and repeating themes. This involved comparing the results of different studies.,

IV. RESULTS

A. Critical Components for Effective Integration of Data-Driven Nudges

The literature review identifies several critical components essential for effectively integrating data-driven nudges. First and foremost is data quality [20]. High data quality is crucial, as inaccuracies, inconsistencies, and incomplete data can lead to unreliable insights and ineffective nudges [21]. To tackle this, robust data governance practices, including data cleaning, validation, and regular audits, are necessary [22]. Additionally, using data integration tools and establishing clear management policies ensures data consistency and reliability [23]. Another key component is understanding the audience. Accurately capturing audience preferences, behaviors, and needs is challenging due to the diverse and dynamic nature of user populations. Misunderstanding these aspects can lead to ineffective or even counterproductive nudges. Comprehensive data collection methods, such as surveys, focus groups, and behavioral analytics, are essential for gaining detailed audience insights [24]. Continuous learning and adaptation using feedback and real-time data help stay aligned with evolving audience preferences [25]. Appropriate AI and ML model development is also crucial. The challenge lies in selecting and developing models that are accurate, interpretable, and applicable to real-world scenarios. Collaboration with skilled data scientists and domain experts is key to achieving this. Ensuring transparency and explainability in model development fosters stakeholder understanding and trust. Regularly updating and retraining models with new data is necessary to maintain their relevance and accuracy. Finally, segmentation and personalization are vital for effective nudges [26]. Obtaining deep insights into individual user characteristics and preferences is challenging, especially at scale, and raises privacy concerns [27]. Advanced analytics and machine learning techniques can identify meaningful user segments, while dynamic content personalization tools tailor messages to resonate with each segment. Ensuring compliance with privacy regulations and maintaining transparency in data usage are essential to maintaining user trust. These components collectively enhance the effectiveness of data-driven nudge strategies.

Segmentation and Personalization	42
Data Quality	39
Appropriate AI and ML Model Development	38
Audience Context	35

Fig. 2. Critical Components for Effective Integration of Data-Driven Nudges

B. Efficient Approaches and Technologies in the Utilization of Data-driven Nudge Strategies

The literature review highlights several innovative approaches and techniques for effectively utilizing data-driven nudge strategies to enhance conversion rates. One key approach is personalized and adaptive nudging, which involves developing systems that continuously learn and adapt based on real-time user data and behavior. This goes beyond traditional methods by using sophisticated algorithms to tailor nudges dynamically, ensuring that each user receives the most relevant nudge at the optimal time [25]. Machine learning models play a crucial role in analyzing user interactions and updating strategies accordingly. Another approach is the integration of behavioral economics with advanced analytics. This combines insights from behavioral economics, such as cognitive biases, loss aversion, and social norms, with advanced data analytics to design more psychologically attuned nudges [28]. Techniques like A/B testing and experimentation frameworks are employed to validate the effectiveness of these nudges, enabling the testing of different message framings and identifying the most impactful strategies [29]. Context-aware nudges are another innovative technique, focusing on the user's current environment, emotional state, or recent actions. This involves integrating context-aware computing and real-time data analytics to gather relevant contextual information [30]. For example, sending reminders or motivational messages at times when users are most likely to engage, based on their past behavior patterns, enhances the relevance and effectiveness of the nudges [31]. Additionally, the review emphasizes the importance of multi-channel and multi-modal nudging, which utilizes a comprehensive approach encompassing digital and physical touchpoints, as well as visual and auditory communication [32]. This strategy ensures that nudge campaigns are delivered consistently across online platforms, mobile apps, SMS, emails, and in-store interactions, creating a cohesive user experience [33]. Lastly, continuous learning and optimization are essential for refining nudge strategies over time [34]. Implementing systems that continuously collect data and use reinforcement learning algorithms allows for the constant improvement of nudges based on user interactions, leading to more effective and engaging interventions [35]. This approach ensures that nudge strategies remain relevant and impactful, thereby significantly enhancing conversion rates.

Personalized and Adaptive Nudging	41
Continuous Learning and Optimization	37
Integration of BE with Advanced Analytics	35
Multi-Channel and Multi-Modal Nudging	31
Context-Aware Nudges	29

Fig. 3. Efficient Approaches and Technique in the Utilization of Data-driven Nudge Strategies

V. DISCUSSION

A. Critical Components for Effective Integration of Data-Driven Nudges

The literature review underscores several critical components for effectively integrating data-driven nudges, each presenting unique challenges and solutions. First, Data Quality is foundational but challenging due to issues like inaccuracies and inconsistencies [36]. These issues can lead to unreliable insights and ineffective nudges. The solution involves implementing robust data governance practices, including data cleaning, validation, and regular audits, along with data integration tools and clear data management policies to maintain high data quality [37]. Understanding the audience is another vital component. The Audience challenge lies in accurately capturing diverse and dynamic user preferences, behaviors, and needs [38]. Misunderstanding the audience can result in poorly designed nudges. Continuous learning and adaptation, driven by feedback and real-time data, are crucial for staying aligned with audience changes. For Appropriate AI and ML Model Development, the complexity of selecting and developing suitable models is a significant challenge, requiring expertise in data science and the specific campaign context [33]. The solution involves collaboration with skilled data scientists and domain experts to develop interpretable and applicable models. Emphasizing transparency and explainability in model development helps stakeholders understand decision-making processes [39]. Regular updates and retraining ensure models remain relevant and accurate. Lastly, Segmentation and Personalization present challenges in obtaining and applying deep insights into individual user characteristics at scale, compounded by privacy concerns [40]. The solution is leveraging advanced analytics and machine learning techniques to identify meaningful segments and using dynamic content personalization tools to deliver tailored messages [41]. Compliance with privacy regulations and transparency in data usage are essential to maintaining user trust [42]. The novelty identified here is the need for an integrated framework that systematically addresses these components. The recommended framework should focus on ensuring high data quality, accurate audience understanding, appropriate AI and ML model development, and effective segmentation and personalization, forming a cohesive strategy for optimizing data-driven nudges.

B. Efficient Approaches and Technologies in the Utilization of Data-driven Nudge Strategies

The discussion from the literature review highlights the promising potential of various innovative approaches and techniques in data-driven nudge strategies, particularly those focusing on personalization, context-awareness, and continuous optimization [43]. However, despite these advancements, there remains a notable gap in the form of a cohesive and comprehensive framework that seamlessly integrates these elements into an optimal nudge strategy. The findings suggest that while personalized and adaptive nudging, integration of behavioral economics with advanced analytics, context-aware nudges, and multi-channel approaches are individually effective, they lack a unified structure that could maximize their collective impact. Currently, the landscape of data-driven nudging is

characterized by fragmented implementations, where each technique operates in silos, potentially limiting the overall effectiveness of the nudge strategies. This fragmentation points to the necessity for a holistic framework that not only consolidates these innovative techniques but also addresses the operational challenges, such as ensuring data quality, ethical considerations, and user privacy [44]. Such a framework would provide a structured methodology for deploying nudges that are not only personalized and contextually relevant but also ethically sound and aligned with user preferences. The recommendation, therefore, is to develop an Optimal Nudge Framework [12]. This framework would integrate advanced machine learning algorithms for real-time personalization, behavioral insights for psychological alignment, context-aware technologies for situational relevance, and multi-channel delivery systems for consistent user engagement. Additionally, it would incorporate continuous learning mechanisms to adapt and optimize nudge strategies based on user feedback and evolving behaviors. By bridging the current gap, this framework would provide a comprehensive solution, enabling more efficient and effective utilization of data-driven nudges, ultimately enhancing conversion rates in digital campaigns and other applications.

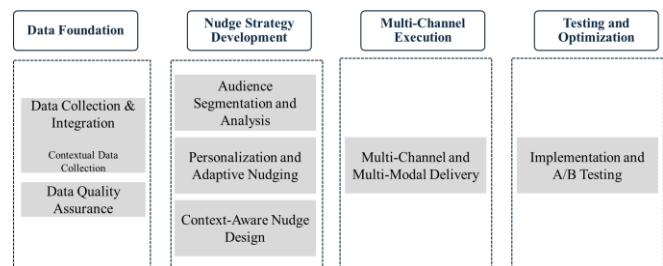


Fig. 4. Data-Driven Nudge Framework

Figure 4 presents the framework for implementing data-driven nudge strategies is categorized into four key areas. The Data Foundation category emphasizes the importance of data collection, integration, and quality assurance, with a focus on gathering contextual data to ensure relevance and accuracy. The Nudge Strategy Development category covers audience segmentation, adaptive nudging, and context-aware design, which are essential for crafting personalized and impactful nudges. The Multi-Channel Execution category highlights the importance of delivering nudges through various channels and modes, ensuring consistent and wide-reaching communication. Finally, the Testing and Optimization category focuses on implementing strategies and employing A/B testing to continuously refine and enhance the effectiveness of the nudges. Each category is crucial in creating a comprehensive and successful nudge strategy framework.

VI. LIMITATION

The following sources were used for this systematic literature review (SLR): ACM Digital Library (dl.acm.org), IEEE Explore (ieeexplore.ieee.org), Springer Link (springerlink.com), and ScienceDirect (sciencedirect.com). Such keywords as “Data Driven Nudges”, “Digital Marketing Campaigns”, “Nudge Theory” and other related terms were used. However, it would be helpful to look at more databases

and use different versions of keywords so that the study is more complete.

VII. CONCLUSION

The application of nudge theory in digital campaigns presents a significant opportunity to enhance conversion rates. However, the existing literature reveals a gap in the comprehensive implementation of these strategies, often due to fragmented approaches and lack of an integrated framework. This study identifies critical components essential for the effective integration of data-driven nudges, including data quality assurance, thorough audience understanding, and the development of appropriate AI and ML models. Furthermore, it emphasizes efficient techniques such as behavioral insights, context-aware nudging, and multi-channel delivery. The novelty of this research lies in the proposed integrated framework that systematically combines these components and techniques, providing a cohesive strategy for optimizing nudge strategies. This framework not only addresses the challenges identified but also offers practical recommendations for enhancing the effectiveness of digital campaigns. By implementing these findings, marketers can better leverage nudge theory, leading to improved user engagement and higher conversion rates, thereby maximizing the impact of their digital campaigns.

REFERENCES

- [1] C. W. Cai, "Nudging the financial market? A review of the nudge theory," *Accounting & Finance*, vol. 60, no. 4, pp. 3341–3365, Mar. 2019, doi: 10.1111/acfi.12471.
- [2] B. Frischmann, "Nudging Humans," *Soc. Epistemol.*, vol. 36, no. 2, pp. 129–152, 2022, doi: 10.1080/02691728.2021.1979121.
- [3] A. H. Sadeghian and A. Otarkhani, "Data-driven digital nudging: a systematic literature review and future agenda," *Behav. Inf. Technol.*, 2023, doi: 10.1080/0144929X.2023.2286535.
- [4] A. Sestino, M. I. Prete, L. Piper, and G. Guido, "The future of online marketing strategies and digital tools: new challenges and contributions to RACE framework," *Int. J. Electron. Trade*, vol. 1, no. 1, pp. 4–27, 2023, doi: 10.1504/IJETRADE.2023.133310.
- [5] V. Fast, D. Schnurr, and M. Wohlfarth, "Regulation of data-driven market power in the digital economy: Business value creation and competitive advantages from big data," *J. Inf. Technol.*, vol. 38, no. 2, pp. 202–229, Jun 2023, doi: 10.1177/02683962221114394.
- [6] A. Makrides, D. Vrontis, and M. Christofi, "The Gold Rush of Digital Marketing: Assessing Prospects of Building Brand Awareness Overseas," *Bus. Perspect. Res.*, vol. 8, no. 1, pp. 4–20, Jan 2020, doi: 10.1177/2278533719860016.
- [7] C. Primasiwi, M. I. Irawan, and R. Ambarwati, "Key Performance Indicators for Influencer Marketing on Instagram," in *Proceedings of the 2nd International Conference on Business and Management of Technology (ICONBMT)*, 2021, doi: 10.2991/aebmr.k.210510.027.
- [8] K. Zieger, U. Schneider, and L. Darcos, "Designing Campaigns and Optimizing Campaign Performance: The Ingredients of Campaign Excellence," *Manag. Prof.*, vol. Part F292, pp. 127–151, 2023, doi: 10.1007/978-3-031-30720-1_8.
- [9] P. Omorogbe, "Improving Digital Marketing Strategy: The Impact of Digital Analytics," 2023.
- [10] M. Poch *et al.*, "Increasing resilience through nudges in the urban water cycle: An integrative conceptual framework to support policy decision-making," *Chemosphere*, vol. 317, no. January, 2023, doi: 10.1016/j.chemosphere.2023.137850.
- [11] M. Franklin, T. Folke, and K. Ruggeri, "Optimising nudges and boosts for financial decisions under uncertainty," *Palgrave Commun.*, vol. 5, no. 1, pp. 1–13, 2019, doi: 10.1057/s41599-019-0321-y.
- [12] F. Callaway, M. D. Hardy, and T. L. Griffiths, "Optimal nudging," in *42nd Annual Meeting of the Cognitive Science Society: Developing a Mind: Learning in Humans, Animals, and Machines*, 2020.
- [13] R. Thaler and S. C., "Nudge: Improving decisions about health, wealth and happiness," *Yale University Press*, 2008.
- [14] C. Almeida, J. Azevedo, A. Fogel, E. Lopes, C. Vale, and P. Padrão, "Effectiveness of nudge interventions to promote fruit and vegetables' selection, purchase, or consumption: A systematic review," *Food Qual. Prefer.*, vol. 116, no. January, 2024, doi: 10.1016/j.foodqual.2024.105122.
- [15] S. Malodia, P. Kaur, P. Ractham, M. Sakashita, and A. Dhir, "Why do people avoid and postpone the use of voice assistants for transactional purposes? A perspective from decision avoidance theory," *J. Bus. Res.*, vol. 146, no. July 2021, pp. 605–618, 2022, doi: 10.1016/j.jbusres.2022.03.045.
- [16] L. T. T. Loan and R. M. Balanay, "Towards reinforcing the waste separation at source for Vietnam's waste management: Insights from the Nudge Theory," *Environ. Challenges*, vol. 10, no. May 2022, p. 100660, 2023, doi: 10.1016/j.envc.2022.100660.
- [17] U. Sivarajah *et al.*, "The effects of business analytics capability on circular economy implementation, resource orchestration capability, and firm performance," *J. Bus. Res.*, vol. 57, no. June, p. 108205, 2021, doi: 10.1016/j.jipe.2021.108205.
- [18] D. Buschmann, T. Schulze, C. Enslin, and R. H. Schmitt, "ScienceDirect Interpretation Framework of Predictive Quality Models for Process- and Product-oriented Decision Support," *Procedia CIRP*, vol. 118, pp. 1066–1071, 2023, doi: 10.1016/j.procir.2023.06.183.
- [19] B. Kitchenham, "Guidelines for performing Systematic Literature Reviews in software engineering," *EBSE Technical Report*, 2007.
- [20] B. M. Kalema and V. V. Busobozi, "Big data analytics for data quality improvement to enhance evidence-based health care in developing countries," in *Pant, M., Sharma, T., Basterrech, S., Banerjee, C. (eds) Performance Management of Integrated Systems and its Applications in Software Engineering*, 2020, doi: 10.1007/978-981-13-8253-6_4.
- [21] A. C. Powell, "Impact of the Artificial Nudge," *Acad. Radiol.*, vol. 27, no. 1, pp. 143–146, 2020, doi: 10.1016/j.acra.2019.09.010.
- [22] X. Zhou, C. Chai, G. Li, and J. Sun, "Database meets artificial intelligence: A survey," *IEEE Transactions on Knowledge and Data Engineering*, vol. 34, no. 3, pp. 1096–1116, March 2022, doi: 10.1109/TKDE.2020.2994641.
- [23] M. K. Saggi and S. Jain, "A survey towards an integration of big data analytics to big insights for value-creation," *Inf. Process. Manag.*, vol. 54, no. 5, pp. 758–790, 2018, doi: 10.1016/j.ipm.2018.01.010.
- [24] S. Motamarri, S. Akter, and V. Yanamandram, "Does big data analytics influence frontline employees in services marketing?," *Bus. Process Manag. J.*, vol. 23, no. 3, pp. 623–644, Jan 2017, doi: 10.1108/BPMJ-12-2015-0182.
- [25] P. Huyn, A. Kumar, and S. Peng, "Personalizing Consumer Interventions Through Leveraging Behavioral Economics," *Procedia Comput. Sci.*, vol. 207, no. Kes, pp. 282–291, 2022, doi: 10.1016/j.procs.2022.09.061.
- [26] R. Karlsen and A. Andersen, "Recommendations with a Nudge," *Technologies*, vol. 7, no. 2, 2019, doi: 10.3390/technologies7020045.
- [27] S. Balkaya, "Role of Trust, Privacy Concerns and Data Governance in Managers' Decision on Adoption of Big Data Systems," *Manag. Stud.*, vol. 7, no. 3, 2019, doi: 10.17265/2328-2185/2019.03.006.
- [28] J. Ifcher and H. Zarghamee, "Behavioral economic phenomena in decision-making for others," *J. Econ. Psychol.*, vol. 77, no. May, p. 102180, 2020, doi: 10.1016/j.joep.2019.06.003.
- [29] R. Kohavi and R. Longbotham, "Online Controlled Experiments and A/B Tests," *Encycl. Mach. Learn. Data Sci.*, pp. 1–13, 2023, doi: 10.1007/978-1-4899-7502-7_891-2.
- [30] M. K. Satheesh and S. Nagaraj, "Applications of artificial intelligence on customer experience and service quality of the banking sector," *International Management Review*, vol. 17, no. 1, pp. 9–17, 2021.
- [31] K. Ruggeri *et al.*, "Nudging New York: Adaptive models and the limits of behavioral interventions to reduce no-shows and health inequalities," *BMC Health Serv. Res.*, vol. 20, no. 1, Apr 2020, doi: 10.1186/S12913-020-05097-6.
- [32] M. Anshari, M. N. Almunawar, S. A. Lim, and A. Al-Mudimigh, "Customer relationship management and big data enabled: Personalization & customization of services," *Appl. Comput. Informatics*, vol. 15, no. 2, pp. 94–101, 2019, doi: 10.1016/j.aci.2018.05.004.
- [33] A. Haleem, M. Javaid, M. Asim Qadri, R. Pratap Singh, and R. Suman, "Artificial intelligence (AI) applications for marketing: A literature-based study," *Int. J. Intell. Networks*, vol. 3, no. July, pp. 119–132, 2022, doi: 10.1016/j.ijin.2022.08.005.
- [34] C. Mele, T. Russo Spena, V. Kaartemo, and M. L. Marzullo, "Smart nudging: How cognitive technologies enable choice architectures for value co-creation," *J. Bus. Res.*, vol. 129, no. March 2019, pp. 949–

- 960, 2021, doi: 10.1016/j.jbusres.2020.09.004.
- [35] S. Mahdavi Hosseini, "Learning framework with joint task optimization applied to consumer health applications with behavioral nudges," *Proc. - 2020 IEEE Int. Conf. Big Data, Big Data 2020*, pp. 5698–5699, 2020, doi: 10.1109/BigData50022.2020.9377938.
 - [36] A. Al-Badi, A. Tarhini, and A. I. Khan, "Exploring big data governance frameworks," *Procedia Comput. Sci.*, vol. 141, pp. 271–277, 2018, doi: 10.1016/j.procs.2018.10.181.
 - [37] V. Gudivada, A. Apon, and J. Ding, "Data quality considerations for big data and machine learning: Going beyond data cleaning and transformations," *International Journal on Advances in Software*, vol. 10, no. 1, pp. 1–20, 2017.
 - [38] Q. Shambour, "A deep learning based algorithm for multi-criteria recommender systems," *Knowledge-Based Syst.*, vol. 211, p. 106545, 2021, doi: 10.1016/j.knosys.2020.106545.
 - [39] P. J. G. Lisboa, S. Saralajew, A. Vellido, R. Fernández-Domenech, and T. Villmann, "The coming of age of interpretable and explainable machine learning models," *Neurocomputing*, vol. 535, pp. 25–39, 2023, doi: 10.1016/j.neucom.2023.02.040.
 - [40] A. Khamaj and A. M. Ali, "Adapting user experience with reinforcement learning: Personalizing interfaces based on user behavior analysis in real-time," *Alexandria Eng. J.*, vol. 95, no. March, pp. 164–173, 2024, doi: 10.1016/j.aej.2024.03.045.
 - [41] C. Zeng, Q. Wang, S. Mokhtari, and T. Li, "Online context-aware recommendation with time varying multi-armed bandit," *Proc. ACM SIGKDD Int. Conf. Knowl. Discov. Data Min.*, vol. 13–17-Aug, pp. 2025–2034, 2016, doi: 10.1145/2939672.2939878.
 - [42] S. Chandra, S. Verma, W. M. Lim, S. Kumar, and N. Donthu, "Personalization in personalized marketing: Trends and ways forward," *Psychol. Mark.*, vol. 39, no. 8, pp. 1529–1562, 2022, doi: 10.1002/mar.21670.
 - [43] S. Dalecke and R. Karlsen, "Designing Dynamic and Personalized Nudges," *ACM Int. Conf. Proceeding Ser.*, vol. Part F1625, pp. 139–148, 2020, doi: 10.1145/3405962.3405975.
 - [44] F. Li and G. Xu, "AI-driven customer relationship management for sustainable enterprise performance," *Sustain. Energy Technol. Assessments*, vol. 52, p. 102103, 2022, doi: 10.1016/j.seta.2022.102103.